Are We Chasing Shadows? Quantifying Unattributable Polarization and Attributing the Rest to Annotator Groups

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Abstract

Polarization has recently been proposed as a robust way of distinguishing minor disagreements from systematic differences in opinion for imbalanced, minority groups but there currently exists no methodology for attributing polarization to these groups. In this study, we discover and investigate two major limitations in realistic settings: (1) the presence of "inherent" polarization that cannot be attributed to any known or latent groups, and (2) opposing polarization effects canceling each other out in aggregated annotations. To address these issues, we introduce a new metric that attributes polarization to any annotator group while avoiding these limitations, and provide an open-source Python library implementation. We apply our method to four subjective NLP datasets and confirm that gender and ethnicity consistently explain polarization patterns, while differences between annotator groups become stronger as the groups are further apart. Our results overall suggest a need for more annotations per item in subjective NLP datasets, although we empirically find that no more than 20 annotators are needed for reliable estimation.

1 Introduction

Annotations are essential for subjective tasks, such as content moderation, toxicity, and HS detection, which are especially important for protecting minority groups (United Nations, 2025; Dioum, 2025). However, in practice analysis rests on inter-annotator agreement or majority-vote (Ivey et al., 2025; Kralj Novak et al., 2022; van der Velden et al., 2025), which may lead to biased and misconfigured systems (e.g., toxicity models disproportionally marking African American (AA) speech as "toxic" (Sap et al., 2019)), or discard

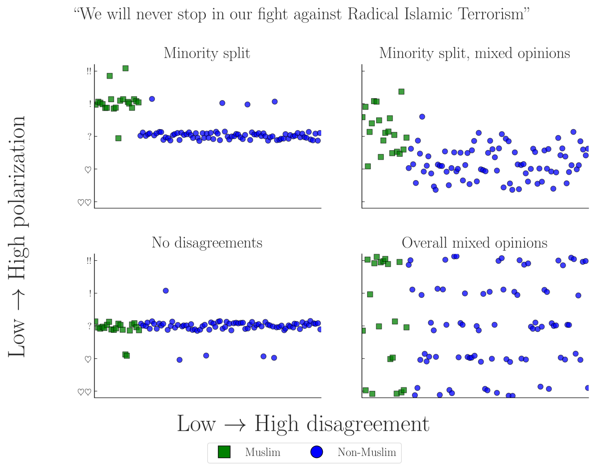

Figure 1: Simulated experiment exploring four scenarios of a Hate Speech (HS) detection task applied to a controversial statement by D. Trump (Jan. 2019). The variance in annotations across different annotators may cause traditional agreement metrics to ignore minority opinions, especially when opinions are mixed (top-right).

fundamental disagreements between minority groups and the majority (Quaranto, 2022), (see the example in Fig. 1).

One alternative is using *polarization*, which correlates better with human judgments and is much less sensitive to different group sizes (Pavlopoulos and Likas, 2023, 2024). While we can detect polarization, there is little work on how to detect whether it is *actually caused by marginalized groups*. Our work thus tries to answer the following Research Question (RQ):

RQ. *How much polarization can we practically attribute to annotator subgroups given existing datasets?*

While prior work assumed that all polarization can be explained and attributed to some annotator characteristic, through the exhaustive creation of theoretical annotator groups our study discovers the existence of

"Inherent Polarization (InPol)", which cannot be attributed to *any possible group* of annotators—known or unknown (Finding 1). We also discover that polarization across comments has a *direction*, preventing us from aggregating annotations on the dataset level to circumvent annotation sparsity (Finding 2). These issues highlight that current subjective Natural Language Processing (NLP) datasets may lack the necessary number of annotators per comment—estimated empirically to be ≈ 20 annotators (§4.4).

In order to solve these issues we introduce a new metric—"*apunim*"—which attributes polarization to specific annotator sociodemographic and ideological characteristics (referred to as Personal Characteristics (PCs) in this study). Our metric is generalizable across any domain and modality (e.g., video/image classification) and, unlike previous approaches, enables both *quantitative* comparisons across datasets and statistical analysis via p-value estimation. Even in cases where unattributable polarization exists, *apunim* is capable of detecting general patterns of polarization across multiple groups.

We then apply our metric to four NLP datasets focused on Toxicity and HS detection (§5), finding that annotator gender and ethnicity significantly impact annotation (Finding 3), and that fundamental disagreements grow more pronounced the further apart annotator groups are (e.g., irreligious vs. very religious annotators—Finding 4). Finally, we release an open source Python (PyPi) library that implements our metric.[1] The code for the experiments, graph creation, and analysis can be found on our project's repository.[2]

Warning: This publication includes examples of controversial and potentially harmful speech for the purposes of demonstration. These examples do not represent the views of the authors.

[1]Library: `https://anonymous.4open.science/r/apunim-5F70`, Webpage: `https://anonymous.4open.science/r/apunim-page`

[2]Replication code: `https://anonymous.4open.science/r/aposteriori-unimodality-E8C1`

2 Background and related work

2.1 Agreement

Popular inter-annotator agreement metrics such as Cohen's kappa, Krippendorff's alpha, and Fleiss's kappa aim to distinguish genuine disagreement from random disagreement through statistical chance-correction. However, these metrics are often difficult to apply to intergroup agreement without extensions or restrictive assumptions (e.g., no missing labels or equal numbers of annotations). They have also been criticized for failing to account for differences arising from social, cultural, and demographic backgrounds (Feinstein and Cicchetti, 1990; Byrt et al., 1993; van der Velden et al., 2025; Checco et al., 2017), producing results that are difficult to compare across studies, being sensitive to class imbalance (Feinstein and Cicchetti, 1990), and relying on questionable assumptions about chance agreement (Checco et al., 2017).

Recent work has addressed some of these limitations through simpler statistical approaches (van der Velden et al., 2025), subgroup-specific adaptations of Cohen's kappa (Wong et al., 2021), and Bayesian models that explicitly separate subgroup disagreement from annotation noise (Ivey et al., 2025). While these methods improve applicability and robustness, they are much more complex and less intuitive compared to their traditional counterparts, indicating that agreement may not be the correct tool for analysis of minority viewpoints.

2.2 Polarization

There are multiple competing formulations for polarization. One of these approaches (Akhtar et al., 2019) utilizes χ^2 to estimate in-group vs out-group variance, but still requires traditional agreement metrics for analysis. Others attempt to solve this problem by framing disagreement as a cluster identification problem, applying either parametric (Checco et al., 2017), or non-parametric (Pavlopoulos and Likas, 2024; Mignemi et al., 2024) approaches to the annotation histograms.

For our work, we use a non-parametric polarization metric, normalized Distance From Unimodality (nDFU), which has been shown to correlate better with human judgment of disagreement than established metrics (Pavlopoulos and Likas, 2024). nDFU takes values in the range $[0, 1]$, where values close to 0 indicate no polarization (i.e., opinions are concentrated around a single mode). The metric is defined as follows:

$$\text{nDFU}(X) = \frac{\max\limits_{i \neq m} \left(f_i - \begin{cases} f_{i-1}, & \text{if } i > m \\ f_{i+1}, & \text{if } i < m \end{cases} \right)}{f_m}$$
$$m = \arg\max_i f_i \tag{1}$$

where, X denotes the set of annotations, f_i the relative frequency of the i-th annotation (e.g., the frequency of 2 in a scale of 1–5), and m the histogram peak. Intuitively, nDFU measures how prominent secondary peaks are compared to the main peak (see Fig. 2 for an example).

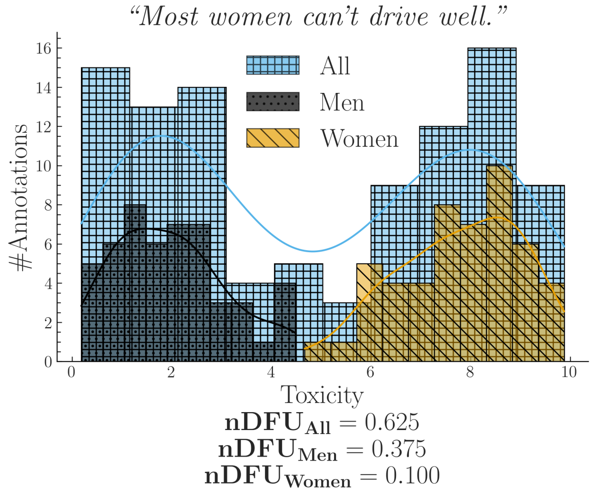

Figure 2: Simulated experiment describing a polarizing comment where male and female annotators agree between themselves, but disagree with the opposite gender.

2.3 Attributing polarization to groups

While many studies study the effect of subgroups on annotation agreement (Goyal et al., 2022; Binns et al., 2017; Giorgi et al., 2025; Al Kuwatly et al., 2020; Sachdeva et al., 2022), attributing *polarization* is a relatively new idea. Under the name *aposteriori unimodality* (Pavlopoulos and Likas, 2024), the original idea is that if polarization in a single item is positive, but zero for all annotator subgroups, then it is "caused" by those groups. Though promising, this mechanism operates on an edge-case scenario where *all* polarization is explained by a single split. Additionally, getting polarization estimates from few annotations per item is extremely noisy, which is especially problematic when the output is binary (i.e., an item either is or is not polarizing) instead of a quantifiable value.[3]

3 Is it possible to attribute all polarization to annotator groups?

3.1 Problem Formulation

Let $d = \{c_1, c_2, \ldots\}$ be a dataset composed of multiple annotated comments.[4] Each comment c is assigned multiple annotators, which creates the annotation multi-set $A(c) = \{(a_1, g_1), (a_2, g_2), \ldots\}$, where a_i is a single annotation (e.g., 2 in a LIKERT scale of 1–5), and g_i a group (e.g., men). As an example, the annotations for a comment in a sentiment analysis task where we group the annotators by gender, would be formulated as:

$$A(\text{"Could be better, could be worse"}) = \{\ (+, F), (-, M), (+, F), \ldots\ \}$$

Intuitively, a group partially explains the polarization of a comment c when the annotations of that group $A(c \mid g) = \{a(c, g') \in A(c) \mid g' = g\}$ exhibit higher polarization compared to the full set of annotations $A(c)$. Consider the case where a misogynistic comment is annotated for toxicity by male and female annotators shown in Fig. 2.[5] The annotations are generally polarized ($nDFU_{all} = .62$); but both female ($nDFU_{women} = .10$) and male ($nDFU_{men} = .37$) annotators exhibit low polarization between their respective groups. This pattern suggests that the overall polarization is partially caused by substantive disagreements between male and female annota-

[3] For example, when presented with noisy estimates of a metric in the $[0, 1]$ range, we would ideally want to distinguish between cases where attribution is close to .001, and cases where attribution is .49.

[4] The annotated items could be of any modality, such images, videos or survey questions, even though we only consider comments in this study.

[5] All simulated experiments in this study were implemented by generating random annotations following the normal distribution with differing means and SD—see https://anonymous.4open.science/r/aposteriori-unimodality-E8C1.

Dataset	#Com	#Ann	#Dims	Task
Kumar	106,035	5	10	Tox
Sap	626	5–6	3	HS
DICES-350	72,103	70–76	4	HS
DICES-990	43,050	123	4	HS

Table 1: Datasets used in this study. *#Ann* refers to the number of annotations per comment (see App. B), and *#Dims* to PC dimensions (e.g., age, gender). Tasks are either Toxicity (Tox) or HS.

tors.[6]

3.2 Datasets

We use four datasets from current NLP research that include PC information at the level of individual annotators; specifically, the "Kumar" (Kumar et al., 2021) and "Sap" (Sap et al., 2022) datasets, which are concerned with Toxicity and HS detection respectively, and the DICES-350 and DICES-990 datasets (Aroyo et al., 2023), which focus on LLM response safety.[7]

A brief overview of dataset characteristics is provided in Table 1. The Kumar dataset stands out for its large size, featuring more than 100,000 annotated comments, as well as 10 distinct PC dimensions.[8] In contrast, the DICES datasets contain fewer annotated comments but involve more than an order of magnitude more annotators per comment than the other datasets. The Kumar and Sap datasets contain a large number of polarizing items, unlike the DICES datasets, where most items are unpolarized (App. B; Fig. 7).[9]

3.3 Inherent Polarization

3.3.1 What is it?

Inherent Polarization (InPol) refers to disagreement that cannot be attributed to *any specific* grouping of annotators; that is, polarization that exists no matter how we split the annotators in any arbitrary group. Prior work (Pavlopoulos and Likas, 2024) attributed polarization to annotator groups only when the observed polarization for each individual group became exactly zero (e.g. $nDFU(all) > nDFU(men) = nDFU(women) = 0$). The presence of InPol would invalidate this theory, since there would be no possible group or combination of groups g *s.t.* $nDFU(g) = 0$.

Definition 1 (Inherent Polarization)**.** Polarization that cannot be explained by any known or latent annotator grouping.

Formally, the InPol for a single comment c is the minimum polarization exhibited by any arbitrary group:

$$Pol_{inh}(c) = \min_{S \subseteq A(c),\, |S| \geq 3} \{nDFU(S)\} \quad (2)$$

When considered exhaustively, these theoretical, random groups include both observed and latent groups (e.g., annotators may be less strict if they had lunch prior to annotation). Since nDFU operates only when a group has at least three annotators (Eq. 1), this yields $\sum_{k=3}^{n} \binom{n}{k}$ possible groups.

3.3.2 Experimental setup

We systematically explore the space of annotator groups by generating random partitions for each comment. Since enumerating all subsets is computationally intractable for large annotator counts, we establish an upper bound via Monte Carlo sampling by drawing k random partitions with random sizes:

$$Pol_{inh}^{MC}(c) = \min_{j=1}^{k} \{nDFU(\tilde{r}_j(A(c)))\} \quad (3)$$

where $\tilde{r}$ is an operator that selects a randomly sized, randomly selected subset. Since Eq. 3 considers a strict subset of Eq. 2, it provides an upper bound for it:

$$Pol_{inh}^{MC}(c) \geq Pol_{inh}(c) \forall c \quad (4)$$

3.3.3 Does it exist?

We compute the InPol for each comment across the four datasets. For Sap and Kumar, we use the exhaustive formulation (Eq. 2), whereas for the two DICES datasets we rely on the Monte Carlo approximation with $k = 1000$ (Eq. 3). Fig. 3 shows that in all DICES comments there exists at least one group that fully

[6] Since polarization persists among male annotators, another subgroup of men may also contribute to it.

[7] For the latter, we consider only labels relating to racism (`Q3_bias_overall`), thereby transforming the task to HS detection.

[8] We use 10,000 comments in this study due to computational constraints—see App. D.3.

[9] This discrepancy could be attributed to these datasets containing LLM-generated dialogues with models tuned to avoid unsafe speech, making it much easier to distinguish than in human comments.

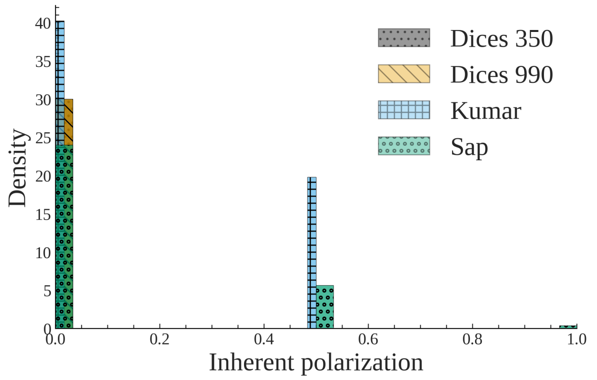

Figure 3: InPol across our datasets (§3.2). Contrary to assumptions made in prior work, InPol is commonly non-zero and can even reach $nDFU = 1$ in datasets with low annotator counts.

accounts for the observed polarization.[10] Interestingly, this observation does not hold for the Kumar and Sap datasets, despite evaluating all possible annotator groups exhaustively. Table 4; App. E shows a few examples of comments with InPol compared to those with none.

Finding 1. *Comments may exhibit a degree of InPol that cannot be attributed to any known or latent annotator grouping given the annotators present in the dataset.*

3.3.4 Why does it exist?

Hypothesis 1 (Low annotator count)**.** Initially, we hypothesized that the absence of InPol in the DICES datasets could be attributed to their large sample size (annotator count per item). However, even when repeating the experiment using random annotator subsets (ranging from 5 to 50 annotators) and taking the maximum across ten random samples for each subset size, we still observed zero InPol (App. E). Thus, sheer annotator count alone is insufficient to explain the phenomenon of InPol.

Hypothesis 2 (Low polarization)**.** We briefly considered the possibility of low overall polarization as an alternative explanation for the absence of InPol in DICES. Nevertheless, it is statistically improbable that this uniformity would hold across all 115,153 comments, given that a substantial fraction do exhibit clear signs of polarization (as shown in Fig. 7; App B).

Hypothesis 3 (Inaccessible groups)**.** InPol may arise from the presence of distinct, partially represented groups within the annotator pool. While these minority groups are responsible for the observed polarization, they may go undetected in our current methodology if they are represented by only one or two annotators, as these comments are discarded during the analysis (§3.3). This is a plausible explanation for the Kumar and Sap datasets but not for DICES; the former experiment should have yielded some InPol if that was the case.

The existence of such polarization in human-text datasets (where the two former explanations apply) suggests an important limitation: current NLP datasets may fundamentally lack the necessary information required for a comprehensive analysis of annotator behavior, particularly in subjective and sensitive domains. We can not offer a convincing explanation for why the DICES datasets present no InPol, even when subsampled to a significant degree, other than structural differences between polarization patterns in human and LLM-generated texts.

3.4 Polarization is directional

One way of potentially overcoming annotator sparsity, is aggregating annotations between comments. However the problem can not be solved that easily; Fig. 4 shows an experiment where we combine annotations across two comments. We observe that, should the two comments be polarized for the opposite reason, the opposing polarization effects might balance each other out, leading to a false negative. Therefore, any statistics must be computed on the comment-level; annotations should never be aggregated across comments.[11]

Finding 2. *Polarization has a unique direction for each comment which may compromise cross-comment aggregation.*

[10] $Pol_{inh}^{MC}(c) = 0 \Rightarrow Pol_{inh}(c) = 0$ due to Eq. 4. The same logic applies if we tried to increase the number of random partitions in Eq. 3.

[11] A similar assumption was followed in early parametric approaches, which assumed different parameters for each comment (Checco et al., 2017).

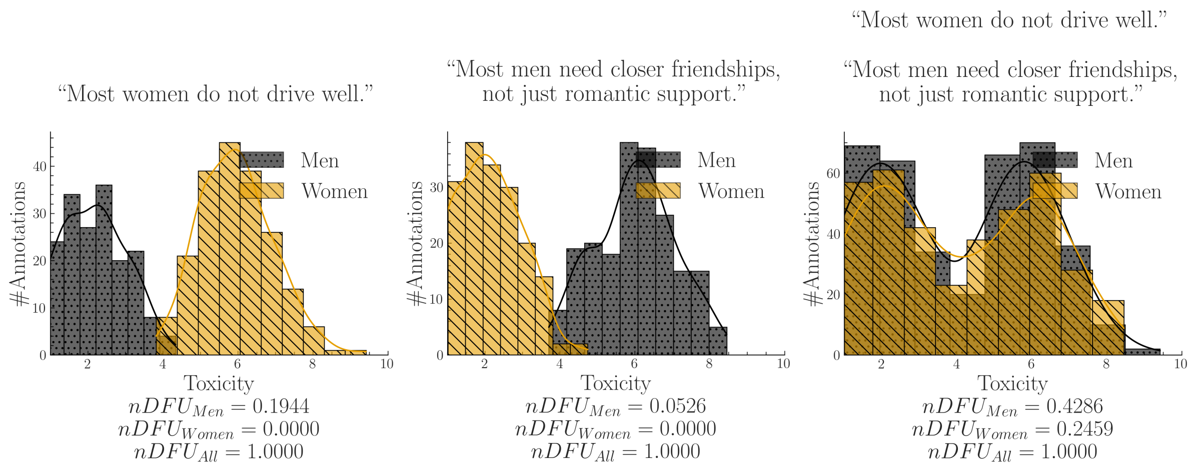

Figure 4: A synthetic experiment simulating a polarizing discussion with two comments where annotators disagree about which one is toxic. When we aggregate the comments, we unintentionally change the comparison: instead of comparing two unimodal distributions (each comment viewed separately) with two bimodal distributions (their aggregated annotations), we end up comparing two bimodal distributions, obscuring the source of the polarization (polarization explained by gender, but not shown).

4 How do we attribute the rest?

4.1 Apriori polarization

Instead of using InPol, which represents the minimum possible polarization exhibited by any group, we calculate the *expected* (average). Therefore, we do not test whether the group is polarized against the whole; nor whether the group explains *some polarization* (as would be the case if we used Eq. 2), but rather whether the group explains polarization *more than other groups*.

Definition 2 (Apriori polarization)**.** The expected polarization of a comment's annotations.

Specifically, we define apriori polarization for a comment c as:

$$Pol_{apr}(c) = \frac{1}{t}\sum_{i=1}^{t} nDFU(\tilde{r}_i(A(c))) \tag{5}$$

where $\tilde{r}$ is the random partition operator (see Eq. 2). We explain why defining apriori polarization in this way is preferable to simpler alternatives in App. C.1.

4.2 Expanding to the whole dataset

We can estimate how much polarization is generally present in the dataset by simply averaging the *apriori* polarization of its comments. Intuitively, a high apriori polarization score means that much of the polarization in the dataset is likely not driven by any single group.

$$Pol_{apr}(d) = \frac{1}{|d|}\sum_{c\in d} Pol_{apr}(c) \tag{6}$$

We compare this quantity with the average observed polarization of all comments:

$$Pol_{obs}(d,g) = \frac{1}{|d|}\sum_{c\in d} nDFU(A(c \mid g)) \tag{7}$$

We can now compare the polarization that is attributed to a single annotator group (Eq. 7) to a prior one (Eq. 6). We define our normalized metric, *apunim*, as:

$$apunim(d,g) = \frac{Pol_{apr}(d) - Pol_{obs}(d,g)}{1 - Pol_{apr}(d)} \tag{8}$$

Definition 3 (Apunim)**.** The degree to which an annotator group contributes to overall polarization in a set of comments.

Since $nDFU \to [0,1] \Rightarrow apunim \to [-1,1]$. Its interpretation boils down to three scenarios:

- $apunim(g) \approx 0$: The examined group does not meaningfully disagree with the other annotators.
- $apunim(g) > 0$: The examined group has fundamental disagreements with the other annotators.
- $apunim(g) < 0$: The annotators of that group are polarized between themselves.

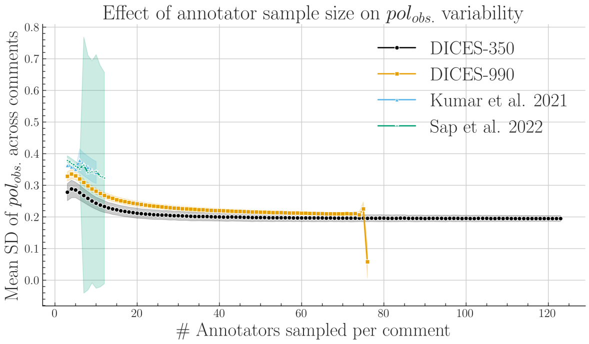

Figure 5: SD of pol_{obs} across comments, averaged across 1000 runs (Eq. 7). Shaded areas represent two standard deviations. Pol. estimates benefit from more annotators up to $n = 20$, at which point they begin to plateau.

4.3 Statistical significance

Since annotation histograms may be noisy, we additionally assess whether the observed apunim value could arise by chance. To assess statistical significance, we construct a null distribution of apunim values from the randomized partitions and compare the observed apunim against this distribution using a one-sample Student-t test. Thus, the null hypothesis states that the observed polarization is indistinguishable from polarization induced by random annotator assignments:

$$H_0 : \ \overline{apunim}_{\text{rand}}(d, g) = apunim(d, g)$$

$$H_a : \ \overline{apunim}_{\text{rand}}(d, g) \neq apunim(d, g)$$

where $\overline{apunim}_{\text{rand}}(d, g)$ denotes the average apunim values obtained from t randomized annotator partitions. For a detailed explanation of the p-value computation and the assumptions underlying the test, refer to App. C.3.

4.4 How many annotators do we need?

We assess the robustness of the polarization statistic $pol_{obs.}$ with respect to the number of annotators per comment using subsampling. For each comment we repeatedly sample n annotators with replacement per comment. We then compute $pol_{obs.}$ and measure the standard deviation of these values across comments. This process is repeated 1000 times and averaged.

Fig. 5 shows a clear, non-linear, inverse relationship between the number of annotators and the standard error of pol_{obs} across all datasets, which can be attributed to better histogram estimation employed by nDFU (§4.1) although the rate of improvement becomes increasingly marginal after 20 annotators. Interestingly, some variability in polarization persists even with an extraordinary amount of annotators, which we hypothesize to be caused by inherent noise in the dataset.

This finding reinforces the fact that most NLP datasets are lacking fine-grained information, as discussed in §3.3.3. However, the indication that only 20 annotators are needed for robust estimations of polarization is encouraging; it is much easier to scale up to this number of annotators compared to creating specializing datasets such as DICES which feature up to 100 annotators. Indeed, our formulation enables us to attribute some polarization even in cases with very low annotator counts, as shown in the next section.

5 Attributing real-life polarization to human groups

5.1 Experimental Setup

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided PC dimensions can partially explain polarization in the annotations. The statistically significant results are presented in Table 2. Note that some groups are not included because there were no valid comments (see App. C.2). See App. D.3 for details on other parameters and execution times, and App. F for the full results.

5.2 Which factors contribute to polarization?

Ethnicity Across all datasets apart from Sap, we observe a consistent pattern in which annotation polarization is linked to annotator ethnicity, and particularly to AA annotators. In fact, *ethnicity is the strongest contributor to polarization* in the DICES datasets (AAs: $-.0428$/Asians: .0659 for DICES-350, all ethnicities for DICES-990) and the second strongest for Kumar (AAs: .2072). These results align with prior work suggesting that ethnic background is a systematic source of annotator disagreement in these tasks (Goyal et al., 2022).

Gender Men and women have statistically significant, opposite effects both in DICES-990

Group	apunim	support
DICES-350		
Race=Asian	-.066	8138
Education=Other	.021	2191
DICES-990		
Age=GenZ	-.063	13741
Age=GenX+	.046	14384
Edu=High school	-.075	7419
Gender=Man	-.027	32494
Gender=Woman	.024	33471
Race=AfricanAm	-.091	5810
Race=Latino	.133	6610
Race=Other	-.074	24321
Race=White	.120	11392
Sap		
Ethnicity=black	.042	4
Gender=Man	-.091	1439
Gender=Woman	.244	877

Table 2: Statistically significant (p¡0.05) apunim results across the four datasets. Gray rows show negative values. Consult App. F for the full results.

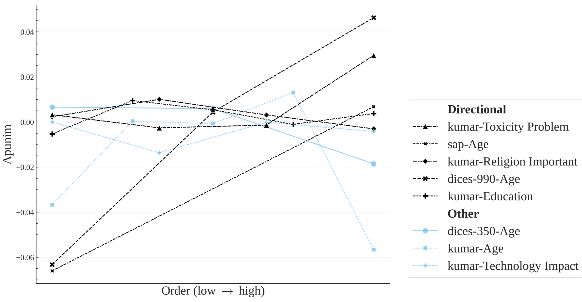

Figure 6: Polarization attribution trends in ordinal groups. The left side of the x-axis refers to young annotators, low education, low religiousness, and negativity towards the impacts of toxicity and technology. The full ordinal modes for each dimension and their individual labels can be found in App. F. We exclude groups with <50 comments, since they near-universally result in zero apunim values due to lack of data.

(Men: .028; Women: $-.025$) and Sap (Men: .091, Women: $-.244$), consistent with previous findings regarding HS (Wojatzki et al., 2018) and toxicity annotation (Binns et al., 2017). *Transgender* annotators also showed the highest degree of polarization in the Kumar dataset (.532).[12]

Finding 3. *Annotator ethnicity and gender are consistent causes of polarization.*

Group distance Fig. 6 shows *apunim* trends across ordinal PC attributes listed in Table 2. We find that *polarization attribution changes consistently* as we move from one end of an attribute scale to the other—e.g., from irreligious to very religious annotators. In some PC dimensions (opinions on toxicity and religion) this is expressed in a monotonic pattern (black lines), suggesting that perceived differences become stronger between groups that are further apart ideologically or demographically. Note that insufficient representation of some groups (see App. F; Table 6) may lead to *apunim* approaching zero due to missing data.

Finding 4. *Polarization attribution tends to grow stronger as the differences between annotator groups increase.*

[12] The fact that a very small group resulted in such a large quantitative attribution is another clue that polarization works well for detecting minority opinions.

6 Conclusion & Future work

We investigated to what extent polarization can be attributed to annotator groups. We discovered as of yet unknown issues regarding the presence of inherent and conflicting polarization, which we treat using a new metric, *apunim*. By applying this metric to four subjective NLP datasets, we discover that (1) ethnicity and gender seem to generally drive polarization attribution, and (2) groups that are further apart increasingly drive polarization. Finally, we estimated the robustness of our metric w.r.t. the number of annotators, and released a PyPi library implementing it.

While we study one group at a time, we can instead turn to iteratively exploring polarized subgroups in future work, allowing us to gradually piece together a multidimensional explanation of polarization (see Footnote 6, §3.1). It would also be interesting to explore to what extent LLM-as-a-judge systems reproduce the polarization patterns observed among different minority groups—which would solve the issues with annotation counts presented in our study.

7 Limitations

Researchers often release only aggregated single-label annotations or omit annotator PC characteristics entirely. As a result, both our analysis and the applicability of our framework are limited by the scarcity of suitable datasets. In addition, because datasets with large numbers of annotators per item are rare, our approach operates only at the dataset level. This limitation prevents us from identifying individual comments or examples that could serve as concrete explanations for the observed *apunim* values.

Finally, although we compare findings across datasets and relate them to prior work, we do not directly compare polarization and agreement metrics, since they rely on fundamentally different assumptions and mechanisms (see §2.2). These differences also make it difficult to fully explain certain patterns in our results, such as why some groups show statistically significant effects while others do not.

8 Ethical Considerations

All datasets used in this study were either publicly accessible, or were provided to us by the authors with full disclosure on the purpose of our research. No identifiable, personal information is included in our pipeline. We did not perform any annotation tasks ourselves.

Our study uses examples of comments which may be considered divisive across certain demographics. These examples were created by us for demonstration purposes—their contents and the groups studied can be replaced with any suitable alternative. We do not claim that these examples necessarily reflect the views or behaviors of these groups, nor are they representative of our own opinions.

We have used small, open source LLMs (primarily LLaMa-8b and Qwen-7B) to help in small style changes throughout this document. We also used them in conjunction with a GPT-5 model to generate some of the code for the plots and experiments presented in this paper. All such outputs have been checked by the authors.

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A Acronyms

InPol Inherent Polarization

nDFU normalized Distance From Unimodality

FWER Family-Wise Error Rate

NLP Natural Language Processing

GPLv3 GNU General Public Licence v3

AA African American

PC Personal Characteristic

HS Hate Speech

B Annotation details

The number of annotators per comment for all datasets presented in §3.2 can be found in 8, and descriptive statistics can be found in Table 3. The DICES-990 dataset has exactly 123 annotations per comment, while the other datasets exhibit a few variations around the

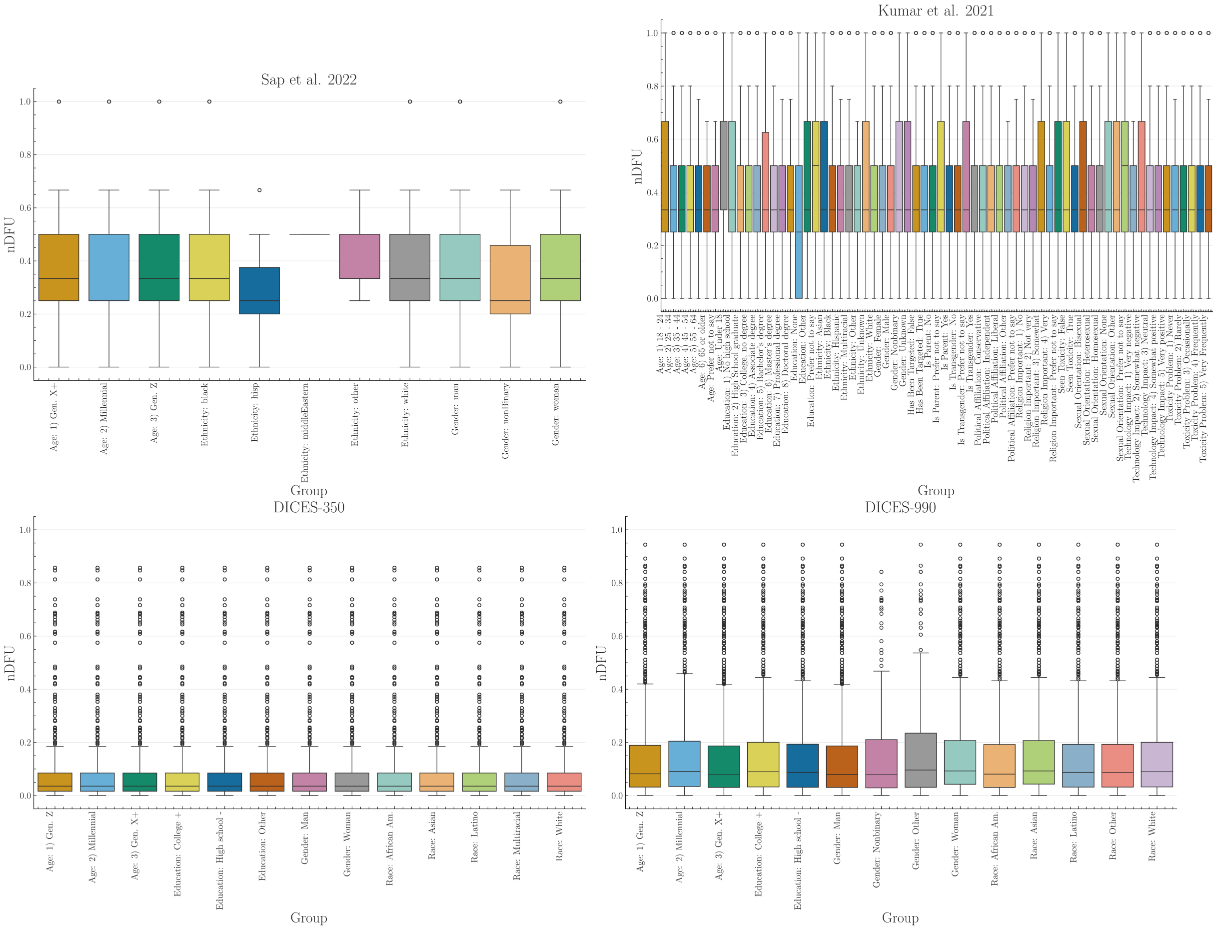

Figure 7: Histogram of dataset polarization for the human datasets considered in this paper per PC dimension.

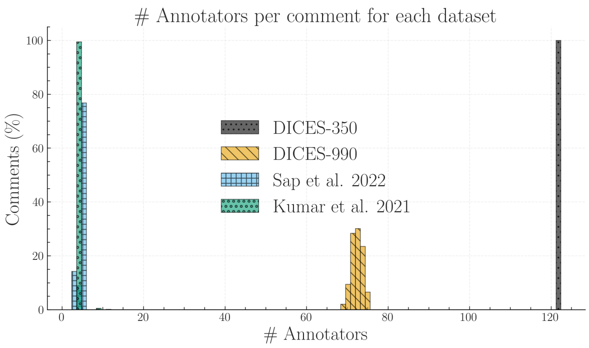

Figure 8: We calculate the ratio of comments for each dataset (y-axis) that had x number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few comments in the Kumar dataset exceeding this threshold (see Table 3), which is likely caused by a data error.

central mean reported in Table 1. Interestingly, the sample used from the Kumar dataset exhibits a few comments that vastly exceed the promised 5 number of annotators. We included these comments for the experiment presented in Table 2, but not for the annotator number experiment in 5 since they are very likely a data error.

C Statistical considerations

C.1 Defining apriori polarization

In §3.1, we explained how our methodology needs to compare the polarization of $A(c)$ and $A(c \mid g)$. Direct comparisons are not advisable, since we would be comparing statistics between a set and its subset—in this case, standard practice dictates comparing the subset with its complement; i.e., the in-group vs. the out-group instead of the in-group vs. all the annotations. However, this also would not work; if we apply the subset-vs-complement method in the example of Fig. 2, we would end up subtracting the nDFUs of men and women, which are similar as they are both unimodal distributions, falsely concluding that there is no polarization present.

Table 3: Descriptive statistics for the number of annotations per comment, grouped by dataset.

	count	mean	std	min	25%	50%	75%	max
DICES-350	350	123	0	123	123	123	123	123
DICES-990	990	72.8313	1.1652	69	72	73	74	76
Kumar et al. 2021	20000	5.0925	4.8872	5	5	5	5	650
Sap et al. 2022	585	5.6359	0.7713	3	6	6	6	12

C.2 Filtering ineligible comments

Because *apunim* is fundamentally designed to quantify polarization, we exclude comments that exhibit no measurable polarization ($nDFU(c) = 0$). Furthermore, we also exclude comments annotated exclusively by a single group (i.e., comments where there are no two groups with at least three annotators each), since we require valid histograms for the non-parametric estimation of nDFU (see Eq. 1). This is an unfortunately common scenario, since many NLP datasets feature only $5-6$ annotations per comment. In a best-case scenario of 5 annotators split evenly between two groups, we would get a 3-2 split.

C.3 P-value estimation

Since our test is parametric (§4.3), it assumes that there exist (1) enough data-points to reliably run the means-test, (2) enough annotations per PC group to estimate nDFUs in Equations 5-7, (3) the apunims of the random partitions are independent and identically distributed. Condition (1) can be safely assumed for most datasets, because they feature enough items to reliably assume normal residuals (see §3.2). We discuss condition (2) in §4.4. Condition (3) is partly fulfilled; the partitions are identically distributed, but may be dependent on the sample distribution.

Replacing the parametric means test with a non-parametric alternative would relax assumptions about the underlying distribution but would not resolve the potential dependence between the random apunims. We initially attempted to implement a permutation test on the empirical (random apunim) distribution, but this resulted in excessive computational costs.

While the test operates on a single group in principle, in practice we often want to test polarization for all subgroups of a specific dimension. Since we are simultaneously considering multiple hypotheses, we apply a multiple comparison correction to the resulting p-values—specifically the Holms method (Holm, 1979). The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses (Chen et al., 2017)). In general, it is safe to set FWER equal to the significance level of our test (e.g., $FWER = 0.95$ if we are looking for $p < 0.05$).

D Software

D.1 Installation and usage

We release the `apunim` package, which is a Python library that implements both the nDFU and apunim calculations, as well as its statistical test. The software is available on PyPi We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (`https://anonymous.4open.science/r/apunim-page`).

The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and can be found on the project's repository (`https://anonymous.4open.science/r/aposteriori-unimodality-E8C1`). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely `numpy, scipy, statsmodels`).

D.2 Implementation details

By default, the computation of the apunim values happens on a single thread. In cases of serious computational cost, we recommend running the apunim calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple PC dimensions, each of which is computationally expensive.

In order to lower the computational cost required, we re-use the random partitions $\tilde{r}_i$), which are originally calculated to obtain the apriori polarization values (see Eq. 5), for the p-value calculations (§4.3). This means that in our current implementation, the number of groups for estimating apriori polarization ($\tilde{r}_i$ in Eq. 5), and the number of apunim values used in the p-value calculation have to be the same.

D.3 Replication Details

We use version `1.0.2` of the `apunim` library for our experiments. We use 100 random iterations for the apriori value calculation (Eq. 5) and the p-value estimation (§4.3), and set a FWER rate of .95.

While the library efficiently handles large numbers of annotators through vectorized operations, performance degrades significantly when processing a high volume of comments, particularly when annotators belong to multiple distinct groups. For instance, while experiments across all datasets in §3.2 complete within a minute, the Kumar dataset necessitates approximately one hour of computation using commercial hardware. Furthermore, the annotator variance analysis detailed in §4.4 required over ten hours. While this latter time requirement is anticipated due to the inherently demanding nature and non-standard application of our library, it demonstrates the existence of a potential scaling bottleneck. The same applies to a lesser extent for the subsampling experiments on the DICES datasets, taking collectively three hours of computation.

We hypothesize that this bottleneck is likely caused by repeated creation of numpy arrays, resulting in repeated memory allocations for each subgroup, in each comment, for each PC dimension. We plan on profiling and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each PC subgroup, or running the different experiments in parallel (indeed, we use the `gnu-parallel` package (Tange, 2025) for our own experiments). We also note that the Kumar dataset also requires substantial RAM resources, estimated around 42–46GB of dedicated RAM should the whole dataset be considered.

Finally, we note that in Fig. 5, we removed values where there was a lack of items annotated by the specified number of annotators (e.g., only 7% of comments in DICES-990 were annotated by more than 73 annotators—see App. B).

E Analyzing InPol further

Table 4 shows a random sample of comments taken from the Sap dataset and their respective InPol values (§3.3). Note that explicitly toxic or racist comments tend to have InPol values of zero, alongside clearly unproblematic comments. Inversely, more ambiguous and less comprehensible comments tend to have larger InPol values. This pattern may suggest that InPol may be tied to the overall ambiguity of the statements, especially in cases where comments may either be interpreted as very toxic, or non-toxic, which agrees with the overall goal of nDFU (§2.2). We include the InPol values for each individual comment for all four datasets in the supplementary material.

In order to investigate whether the numerous annotators per comments can explain the lack of InPol in the DICES datasets (§3.3.4) we gradually subsample the annotators for each comment randomly with replacement, starting from five annotators per comment, and scaling up to the full complement of annotators in the dataset with a step of two. For each such step, we choose ten annotator samples and compute the mean InPol across all runs. Fig. 9 demonstrates that even with annotator counts similar to the Kumar and DICES datasets, InPol remains zero, suggesting that the absence of InPol can not be attributed to annotator number alone—rather, it must be a pattern attributable to the DICES datasets themselves.

F Full Results

Comment	Inh. Pol.
The minks mentioned in the headline are indeed ”rising” from where they were based, but that is due to them being buried in shallow mass graves (Africa) and filling with gas that causes them to rise out. Rest assured, they’re still very much dead, thankfully!.	1
There should be such thing as a bitch resistance for us women	1
There’s a big difference between being a bitch and being a woman	1
i just watched a video of @joisnotonfire being an annoying hoe on skype i’m laughing	1
niggas love to commit crime	1
I swear to God above I love ya and when ya be with this brotha, I swear I love you my bro as long as you not a nigga	0.5
I will become a cold ass bitch because of my treatment	0.5
I would happily lay out a nigga	0.5
Shoot yourself in the head you hypersensitive fuckin bitch.	0.5
not all women are bitches, but all bitches are women	0.5
“ you ugly bitch!!!”- that seems to be my real name these days”	0
“I am happy to report that drowning does kill this demonic entity. Oh holy are the waters of Chernarus!”” Omg, DED ;)”	0
“Let them praise the name of the LORD: for his name alone is excellent; his glory is above the earth and heaven.”” Psalms 148:13 #Bible”	0
someone plz get this nigger some help. He’s playing up like mad	0
stop talking you fucking nigger !	0

Table 4: Sample of comments from the Sap dataset and their InPol. Note that unambiguously toxic comments can be assigned with zero InPol, while more nuanced cases tend to have higher values.

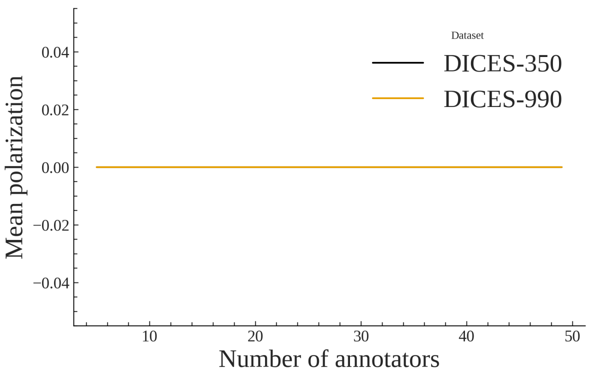

Figure 9: InPol for both DICES datasets averaged across ten runs, subsampling annotators starting from five to the actual number of annotators with a step of two. InPol remains zero, no matter how many annotators per comment are considered.

Group	apunim	pvalue	support
	Sap		
Age=GenX+	-.066	0.708	271
Age=Millennial	.007	0.744	1886
Ethnicity=black	.042	4.000	4
Ethnicity=white	-.000	0.998	1896
Gender=Man	-.091	0.000	1439
Gender=Woman	.244	0.000	877

Table 5: Apunim results for the Sap Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.

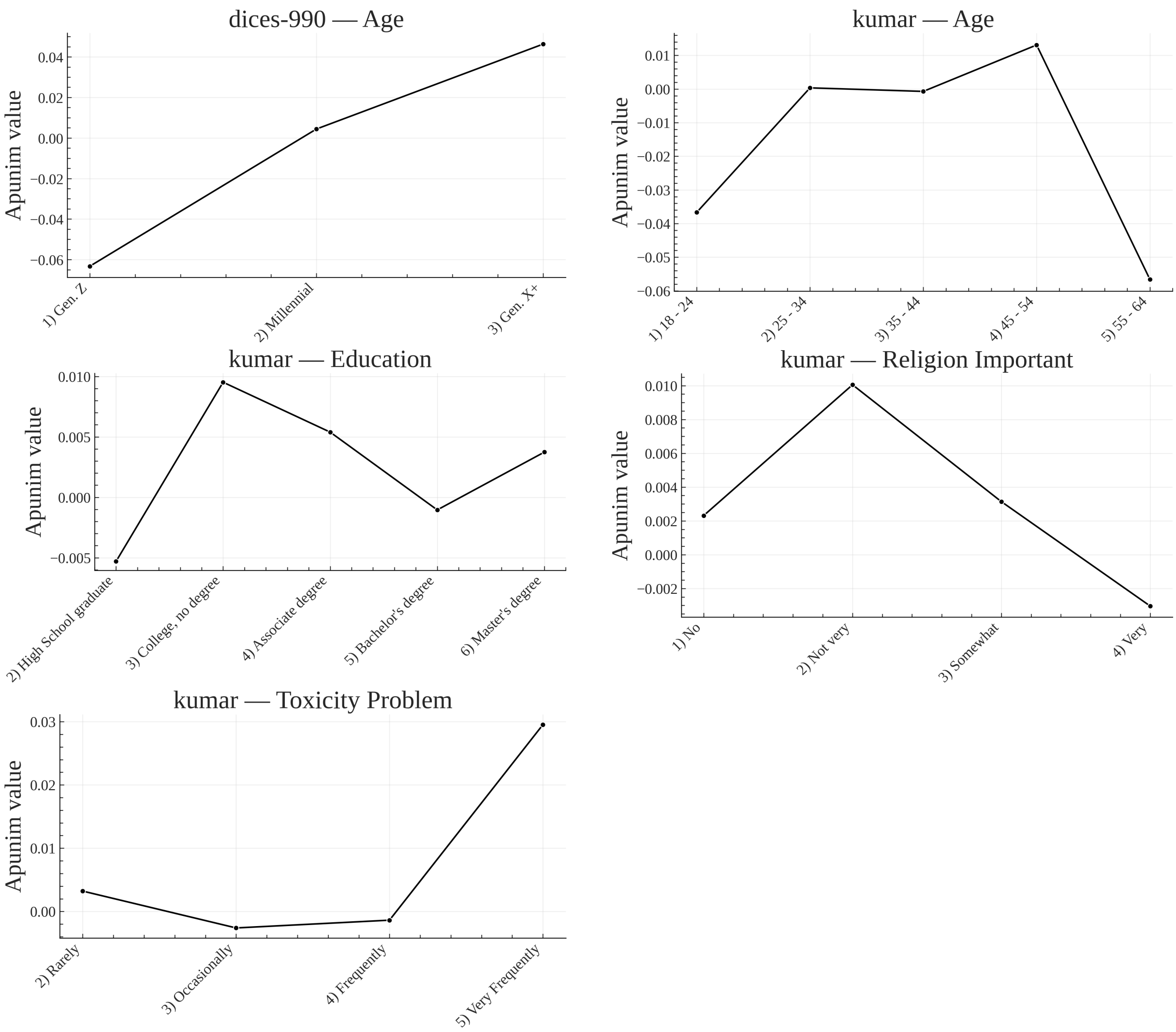

Figure 10: Apunim values across ordinal factor levels with at least two statistically significant values, expanded for each ordinal PC dimension. This figure is complementary to Fig. 6. The values in the graphs were extracted from the Tables of F.

Group	apunim	pvalue	support
Kumar			
Age=1) 18-24	-.037	0.994	308
Age=2) 25-34	.000	0.994	8349
Age=3) 35-44	-.001	0.994	2485
Age=4) 45-54	.013	0.994	410
Age=5) 55-64	-.057	0.994	91
Education 2) High School graduate	-.005	0.952	139
Education 3) College, no degree	.010	0.952	1379
Education 4) Associate degree	.005	0.952	310
Education 5) Bachelor's degree	-.001	0.952	8856
Education 6) Master's degree	.004	0.952	794
Ethnicity=Asian	.011	0.996	39
Ethnicity=Black	-.004	0.996	431
Ethnicity=Hispanic	.000	0.996	9
Ethnicity=Multiracial	.000	0.996	3
Ethnicity=White	-.000	0.996	26007
Gender=Female	.002	0.862	14290
Gender=Male	-.002	0.862	11566
Has Been Targeted False	-.000	1.000	25541
Has Been Targeted True	-.003	1.000	3785
Is Parent No	.001	0.993	10627
Is Parent Prefer not to say	1.000	0.993	3
Is Parent Yes	-.000	0.993	15749
Is Transgender No	.000	0.995	40079
Political Affiliation Conservative	-.003	0.944	2983
Political Affiliation Independent	.001	0.944	3183
Political Affiliation Liberal	.001	0.944	7765
Political Affiliation Other	.000	0.944	18
Political Affiliation Prefer not to say	.000	0.944	12
Religion Important 1) No	.002	0.884	3957
Religion Important 2) Not very	.010	0.884	286
Religion Important 3) Somewhat	.003	0.884	2450
Religion Important 4) Very	-.003	0.884	5292
Seen Toxicity False	-.001	0.980	2384
Seen Toxicity True	.001	0.980	28556
Sexual Orientation Bisexual	-.040	1.000	438
Sexual Orientation Heterosexual	-.000	1.000	31922
Sexual Orientation Homosexual	.000	1.000	6
Technology Impact 2) Somewhat negative	.000	0.999	186
Technology Impact 3) Neutral	-.014	0.999	392
Technology Impact 4) Somewhat positive	-.000	0.999	13500
Technology Impact 5) Very positive	-.004	0.999	2846

Group	apunim	pvalue	support
DICES-350			
Age 1) Gen. Z	0.007	0.539	17528
Age 2) Millennial	0.006	0.539	11268
Age 3) Gen. X+	-0.019	0.166	9703
Education College +	-0.012	0.234	23475
Education High school	-0.015	0.186	12833
Education Other	0.021	0.025	2191
Gender Man	0.019	0.058	19093
Gender Woman	-0.021	0.058	19406
Race African Am.	0.043	2.41e-05	9077
Race Asian	-0.066	1.16e-10	8138
Race Latino	0.003	0.920	6886
Race Multiracial	0.000	0.980	5008
Race White	0.020	0.067	9390

Table 7: Apunim results for the DICES-350 Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.

Group	apunim	pvalue	support
DICES-990			
Age 1) Gen. Z	-0.063	3.05e-25	13741
Age 2) Millennial	0.004	0.495	38436
Age 3) Gen. X+	0.046	7.17e-15	14384
Education College +	-0.009	0.163	59142
Education High school	-0.075	5.80e-42	7419
Gender Man	-0.027	3.81e-05	32494
Gender Woman	0.025	0.00015	33471
Race African Am.	-0.091	1.61e-64	5810
Race Asian	-0.011	0.080	18428
Race Latino	0.133	7.57e-112	6610
Race Other	-0.075	1.51e-30	24321
Race White	0.120	6.03e-81	11392

Table 8: Apunim results for the DICES-990 Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.